

GRAND JUNCTION, CO, USA

WMO: 724760

Lat: 39.1200N

Lon: 108.5250W

Elev: 1473

StdP: 84.84

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23066

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-11.7	-19.4	0.8	-8.5	-17.2	1.0	-4.7	8.9	3.7	7.9	1.4	1.8	80	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	36.6	16.2	35.3	15.7	34.0	15.4	18.5	29.6	17.9	29.0	17.3	28.5	4.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	13.7	20.0	14.9	12.7	20.3	13.6	11.6	21.1	58.1	29.1	56.0	28.8	54.0	28.4	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	8.8	7.7	-17.6	38.7	3.9	1.0	-20.4	39.5	-22.7	40.0	-24.9	40.6	-27.7	41.3		
			-18.1	19.7	3.6	0.8	-20.8	20.3	-22.9	20.8	-24.9	21.3	-27.6	21.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-1.9	2.0	7.3	11.1	16.6	23.0	26.5	24.7	19.6	11.8	4.5	-1.7	(6)
(7)		DBStd	10.52	4.58	4.19	4.18	4.33	4.44	3.75	2.46	2.32	3.92	4.31	4.33	4.73	(7)
(8)		HDD10.0	1304	368	225	104	39	4	0	0	1	31	169	363	(8)	
(9)		HDD18.3	3009	626	458	342	218	86	9	0	31	205	414	621	(9)	
(10)		CDD10.0	2040	0	1	21	73	209	391	511	454	289	87	5	0	(10)
(11)		CDD18.3	703	0	0	0	1	32	149	253	196	69	3	0	0	(11)
(12)		CDH23.3	8822	0	0	3	44	514	2039	3174	2194	802	51	0	0	(12)
(13)		CDH26.7	4378	0	0	0	4	161	1055	1769	1086	297	6	0	0	(13)
(14)	Wind	WSAvg	3.4	2.3	2.8	3.5	4.0	4.0	4.2	4.0	3.8	3.7	3.2	2.6	2.2	(14)
(15)	Precipitation	PrecAvg	225	15	13	21	22	22	11	16	23	26	25	17	15	(15)
(16)		PrecMax	377	42	34	51	58	52	53	49	80	79	88	51	52	(16)
(17)		PrecMin	114	0	0	0	2	0	0	0	2	0	0	0	0	(17)
(18)		PrecStd	58	9	9	14	14	15	13	13	15	21	19	11	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	16.4	23.4	26.8	33.2	37.5	38.5	36.7	34.0	27.3	19.1	12.2	(19)
(20)		MCWB	4.7	6.9	9.2	10.9	14.0	15.5	16.9	16.4	15.8	12.4	8.6	5.5	(20)	
(21)		2%	DB	8.6	13.3	20.6	24.5	30.3	35.6	37.2	35.3	31.9	24.6	16.9	9.1	(21)
(22)		MCWB	3.3	5.2	8.0	9.7	12.7	14.7	16.7	16.0	14.9	11.3	7.2	3.9	(22)	
(23)		5%	DB	6.5	11.0	18.2	22.6	28.3	34.2	35.9	33.9	30.1	22.3	14.6	7.0	(23)
(24)		MCWB	2.3	4.3	7.0	9.0	12.0	14.3	16.3	15.7	14.0	10.5	6.3	2.5	(24)	
(25)		10%	DB	4.6	8.7	15.9	20.4	26.4	32.6	34.6	32.4	28.3	20.0	12.1	4.9	(25)
(26)		MCWB	1.2	3.2	6.0	8.3	11.4	13.9	16.1	15.7	13.5	9.4	5.4	1.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.7	10.0	11.8	15.7	17.3	19.4	19.1	17.6	14.3	9.6	6.5	(27)
(28)		MADB	8.7	14.2	21.0	24.3	29.4	32.0	30.4	28.1	27.7	21.4	17.5	10.7	(28)	
(29)		2%	WB	4.1	6.0	8.6	10.5	13.9	16.3	18.6	18.4	16.6	12.8	8.0	4.7	(29)
(30)		MADB	7.3	11.5	19.0	21.9	26.3	30.5	30.6	28.4	25.9	21.1	14.9	8.4	(30)	
(31)		5%	WB	2.8	5.0	7.6	9.7	12.9	15.5	18.0	17.8	15.8	11.7	6.9	2.9	(31)
(32)		MADB	5.9	9.8	16.4	20.4	25.1	30.1	30.0	27.7	24.8	19.3	13.2	6.1	(32)	
(33)		10%	WB	1.5	3.9	6.5	8.8	12.0	14.8	17.5	17.2	15.0	10.5	5.7	1.6	(33)
(34)		MADB	4.0	8.0	14.8	19.0	24.3	29.3	29.6	27.2	23.9	18.2	11.4	4.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.2	11.0	13.5	13.8	14.9	16.4	15.5	14.8	14.6	13.7	12.3	10.1	(35)	
(36)		5% DB	MCDBR	11.9	14.0	16.7	17.1	17.5	18.0	17.2	16.9	16.8	16.4	15.3	11.9	(36)
(37)		MCWBR	7.4	7.6	8.0	7.5	7.0	6.8	6.1	6.1	6.6	7.6	8.3	7.4	(37)	
(38)		5% WB	MCDBR	10.0	12.2	15.3	15.4	16.1	16.2	14.5	13.3	12.9	13.4	13.6	10.5	(38)
(39)		MCWBR	6.5	6.9	7.5	7.0	6.4	6.4	4.9	4.6	5.2	6.4	7.5	6.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.260	0.296	0.315	0.327	0.323	0.368	0.372	0.321	0.297	0.267	0.256	(40)	
(41)		taud	2.561	2.569	2.466	2.438	2.457	2.469	2.389	2.377	2.514	2.547	2.573	2.548	(41)	
(42)		Ebn at Noon	943	983	967	962	951	950	904	893	928	922	919	915	(42)	
(43)		Edh at Noon	71	81	100	110	111	110	118	116	95	82	70	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.58	3.39	4.76	5.98	6.87	7.93	7.11	6.27	5.46	4.00	2.84	2.19	(44)	
(45)		RadStd	0.16	0.26	0.36	0.36	0.46	0.46	0.36	0.28	0.28	0.32	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	-1.48	-0.91	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page